

Fun with Numbers!

**Percentages —
Reverse Percentage &
Error**



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What Are Percentages?

A percentage is a way to express a number as a part of 100. For example, 50% means 50 out of 100. Think of it like slicing a pizza into 100 pieces!

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Understanding Reverse Percentage

Reverse percentage helps us find the original number before a percentage was added or taken away.

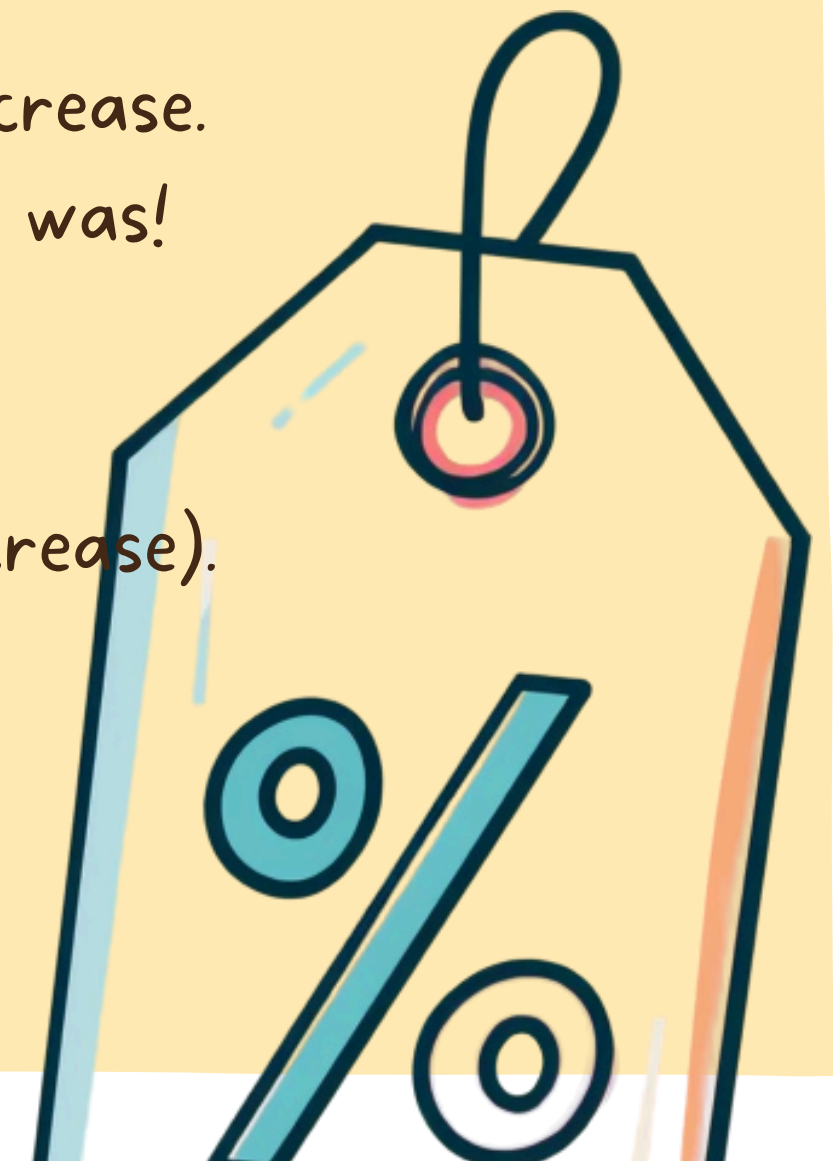
Imagine you know the final price of something after a percentage increase or decrease. Reverse percentage lets you work backwards to discover what the original price was!

Example:

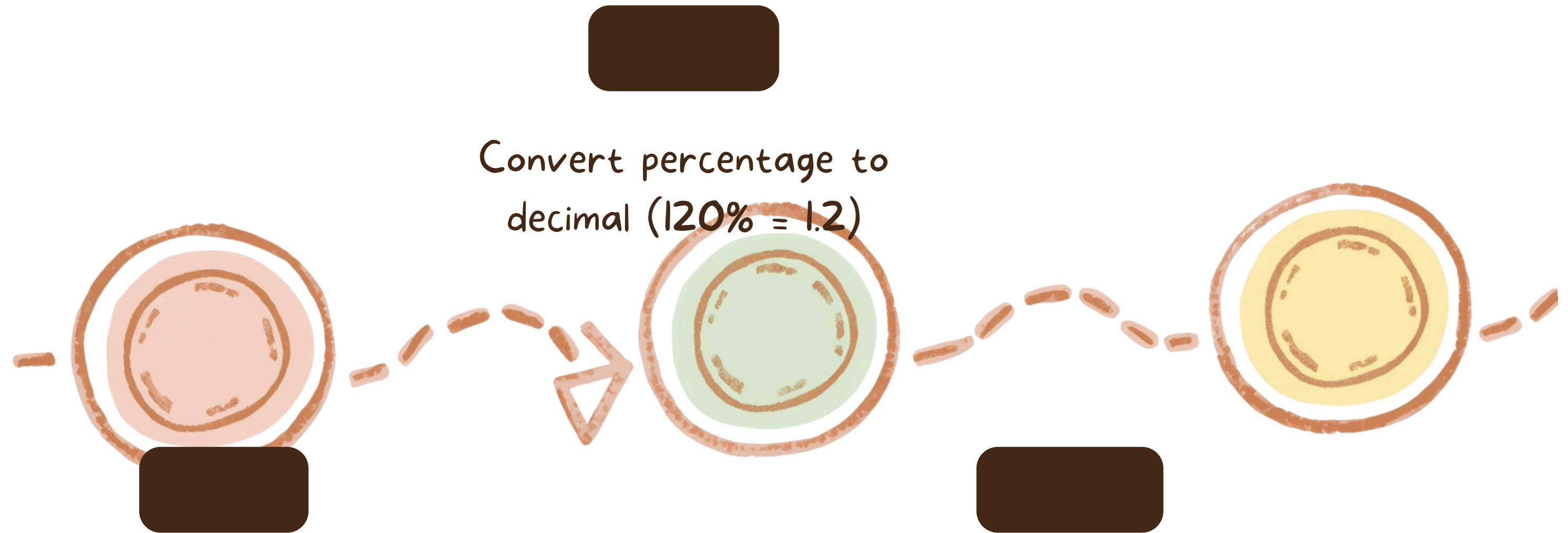
A shirt costs ₹120 after a 20% increase. What was the original price?

Think about it: The ₹120 represents 120% of the original price (100% + 20% increase).

Now we need to find what 100% was!



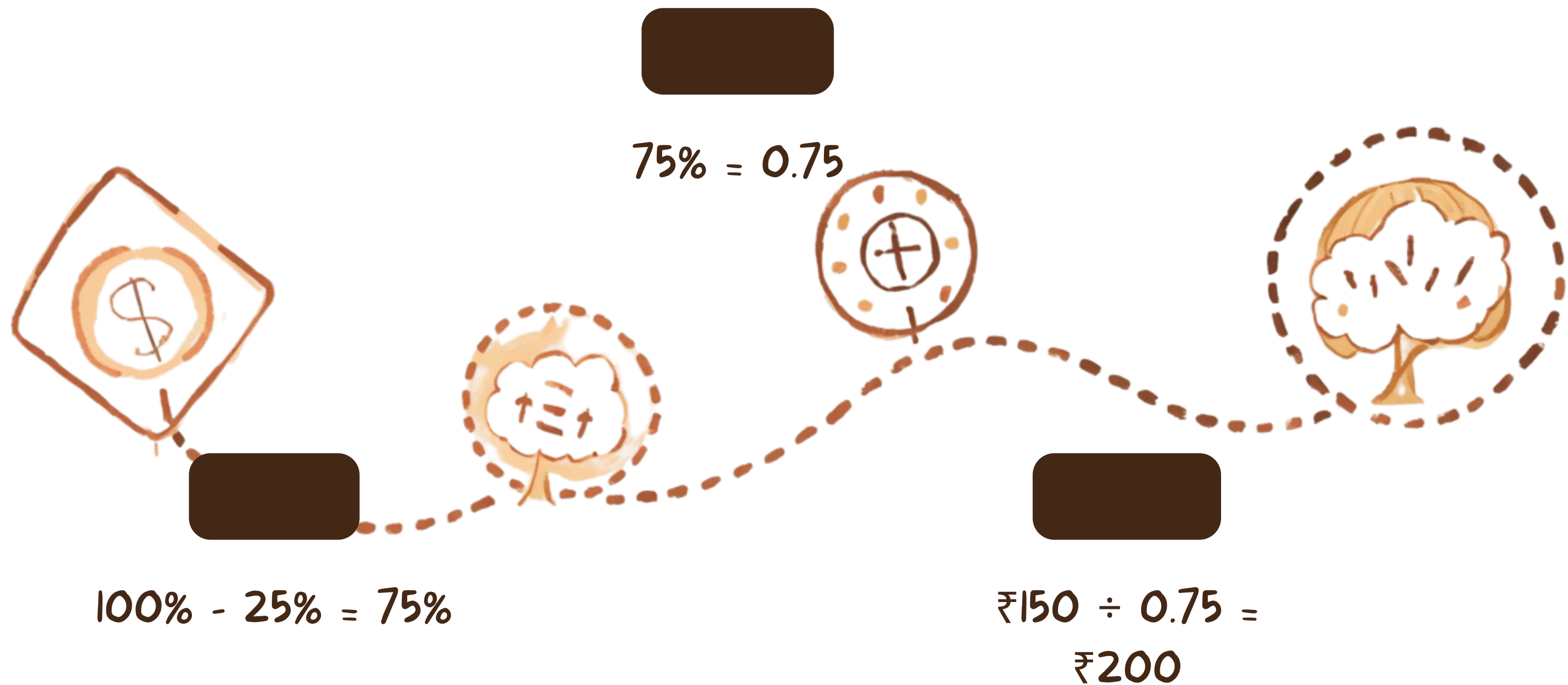
How to Solve Reverse Percentage Problems



Add 100% to the given percentage increase (e.g., $100\% + 20\% = 120\%$)

Divide the final price by the decimal to find original price ($\text{₹}120 \div 1.2 = \text{₹}100$)

Reverse Percentage Example



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Practice Reverse Percentage



Try This Problem!

A bicycle costs ₹1,800 after a 10% discount. What was the original price? Think step by step!

Remember the Steps

Find the percentage remaining ($100\% - \text{discount}\%$).
Convert to decimal. Divide the sale price by the decimal to find the original!

Check Your Answer

Hint: $100\% - 10\% = 90\% = 0.90$. Now divide ₹1,800 by 0.90. Did you get ₹2,000? Great job!

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Solution: Practice Problem

A bicycle costs ₹1,800 after a 10% discount.

Step 1: $100\% - 10\% = 90\%$

Step 2: $90\% = 0.90$

Step 3: $₹1,800 \div 0.90 = ₹2,000$

The original price was ₹2,000!





What Is Percentage Error?

Percentage error tells us how far off our estimate is from the actual value.

It helps us measure how accurate our guess or measurement was.

The smaller the error, the better our estimate!

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Percentage Error Formula

Percentage Error =

$$(|\text{Estimated Value} - \text{Actual Value}| \div \text{Actual Value}) \times 100\%$$

Key Points:

- Always use absolute value (ignore negative signs)
- Always divide by the actual value
- Multiply by 100 to get the percentage





Worked Example: Percentage Error

Riya estimated 45 mangoes in a box.
Actual count was 50.

$$\begin{aligned}\text{Error} &= |45 - 50| \div 50 \times 100\% \\ &= 5 \div 50 \times 100\% \\ &= 10\%\end{aligned}$$

Riya's estimate was 10% off!

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Practice: Percentage Error

Try this problem:

Arjun guessed the length of a rope as 8 metres.
The actual length is 7.5 metres.

Find the percentage error.

Hint: Use the formula!

$|\text{Estimated} - \text{Actual}| \div \text{Actual} \times 100\%$





Solution: Percentage Error

$$\begin{aligned} &|8 - 7.5| \div 7.5 \times 100\% \\ &= 0.5 \div 7.5 \times 100\% \\ &= 6.67\% \text{ (approx)} \end{aligned}$$

Arjun's estimate was about
6.67% off from the actual length.

Well done if you got it!

More Practice Problems

1. A watch costs ₹450 after a 10% increase.
Find the original price.
2. Priya estimated 60 students in a hall.
Actual count was 55. Find % error.
3. A bag costs ₹680 after a 15% discount.
What was the original price?



Key Takeaways

- Reverse percentage finds the original value before a % change
- Always work with 100% as your base
- Percentage error measures accuracy of estimates
- Formula: $|\text{Est} - \text{Actual}| \div \text{Actual} \times 100\%$
- Practice makes perfect!

Quick Quiz!

1. A toy costs ₹180 after a 20% discount.
What was the original price?
2. Estimated weight: 52 kg. Actual: 50 kg.
Find the percentage error.
3. Price after 15% increase is ₹690.
Find the original price.



Thank You!

**Keep practising percentages!
You're doing amazing.**

**Remember: Maths is fun
when you try!**